

|                                                                                   |                                                                                                                                                                                |                               |                                   |
|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------|-----------------------------------|
|  | <b>Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded</b><br><b>Department of CSE</b><br><b>End Term</b><br><b>Academic Year: 2025-26 (Semester – I)</b> |                               |                                   |
|                                                                                   | <b>Class:</b><br><b>SY-B.Tech</b>                                                                                                                                              | <b>Subject: Number Theory</b> |                                   |
|                                                                                   | <b>Date: 28.11.2025</b>                                                                                                                                                        | <b>Time: 13:30 to 15:30</b>   |                                   |
|                                                                                   |                                                                                                                                                                                |                               | <b>Subject Code: MDM-IS-CS201</b> |
|                                                                                   |                                                                                                                                                                                |                               | <b>Max. Marks: 60</b>             |

**Instructions:** i. Attempt All questions.  
ii. Use of a non-programmable scientific calculator is allowed.  
iii. Write clear Steps and justify results where required.

|     |                                                                                                                                                                                                                                          |    |              |     |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|--------------|-----|
| Q.1 | Find all integer solutions to the Diophantine equation: $35x+50y=5$<br>(Show GCD computation + general solution )                                                                                                                        | 10 | CO 1<br>CO 3 | BT1 |
| Q.2 | Solve the CRT system :<br>$x \equiv 4(mod 7), x \equiv 2(mod 9) , x \equiv 5(mod 10)$<br>Show clear CRT steps:<br>computing $M_i$ , inverses, and final combination.                                                                     | 6  | CO 5         | BT5 |
| Q.3 | Day of week of <b>26 January 1950</b> . And <b>28 November 2025</b> by using <b>Zeller's Congruence</b>                                                                                                                                  | 6  | CO 2         | BT2 |
| Q.4 | Compute $ord_9(2)$ and $ord_{14}(3)$ .                                                                                                                                                                                                   | 5  | CO 2         | BT3 |
| Q.5 | Find $\tau(n)$ and $\sigma(n)$ for $n = 60$ .                                                                                                                                                                                            | 5  | CO 4         | BT5 |
| Q.6 | Find $\phi(n)$ for $n = 12, 18, 39, 45, 100$ , and verify Euler's theorem for each.                                                                                                                                                      | 10 | CO 1         | BT6 |
| Q.7 | Using the element <b><math>3 \pmod 7</math></b> , determine its order , check whether it is a primitive root and compute all quadratic residues <b><math>\pmod{17}</math></b> , explain each step clearly with the required definitions. | 10 | CO 4         | BT3 |
| Q.8 | a) Use the Euclidean Algorithm to compute <b><math>\gcd(414, 252)</math></b> .<br>b) Prove by induction: <b><math>1 + 3 + 5 + \dots + (2n - 1) = n^2</math></b> .                                                                        | 8  | CO 3<br>CO 5 | BT4 |

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

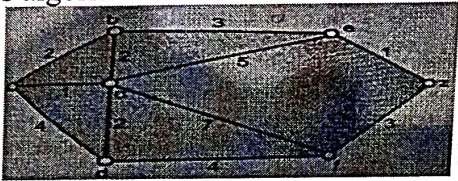
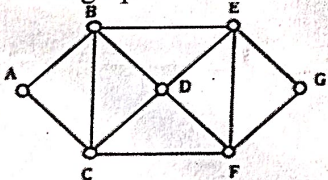
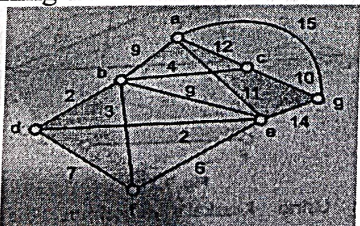
|      |                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------|
|      | On successful completion of this course, students should be able to:                                                |
| CO 1 | Understand all concepts of congruencies and modular arithmetic                                                      |
| CO 2 | Apply modular arithmetic to real-world problems                                                                     |
| CO 3 | Apply the Euclidean algorithm to compute greatest common divisors (GCDs)<br>And express GCD as linear combinations. |
| CO 4 | Model and solve number-theoretic problems relevant to cryptography and Algorithm design.                            |
| CO 5 | Analyze and evaluate problem solving strategies in elementary number theory<br>And communicate solutions clearly.   |



|                  |                                   |                        |
|------------------|-----------------------------------|------------------------|
| Class: SY        | Course Name: Discrete Mathematics | Course Code: PCC-CS201 |
| Date: 26/11/2025 | Time: 1:30 to 3:30                | Max. Marks: 60         |

**Note:**

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Figures to right indicate full marks.
- iii. All questions are compulsory.

| Q.N. |                                                                                                                                                                                                                                                | Marks | CO  | BT       |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|----------|
| Q.1  | Explain the concept of Integral Domain and Field with example. Show that the group $G$ is cyclic group and find all generators, where $G = \{1, 3, 5, 7\}$ for $(G, +_8)$ .                                                                    | 10    | CO3 | BT3      |
| Q.2  | What is group? State and prove the necessary and sufficient condition for a non-empty subset $H$ of a group $Z$ to be a subgroup. Where $(Z, +)$ is a group and $H = \{1, 2, 3, 4, \dots\} \subseteq Z$ .                                      | 10    | CO4 | BT3      |
| Q.3  | Define a complete graph, Bipartite graph, Regular graph, Planar graph and Connected graph. [Each carry 2 marks]                                                                                                                                | 10    | CO5 | BT1      |
| Q.4  | Find the shortest path between $a$ to $z$ in the given weighted graph using dijkstra's algorithm.<br>                                                       | 6     | CO5 | BT3      |
| Q5   | Does the following graph has Hamiltonian circuit and Euler circuit? If exist draw such graphs.<br>                                                          | 4     | CO5 | BT4      |
| Q.6  | Define Tree with example. Construct Binary Search Tree using the keys – 40,20,10,25,60,50,65,70,30<br>Then find Inorder, Preorder, Postorder Traversal. Also find the height of the Tree.                                                      | 10    | CO5 | BT2, BT6 |
| Q.7  | What is Spanning tree? Find the minimum spanning tree for the graph shown below using prim's algorithm. Obtain the total cost of minimum spanning tree.<br> | 10    | CO5 | BT2, BT4 |

== ALL THE BEST ==

**Bloom's Taxonomy (BT)**

**BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create**

**Course Outcomes:**

|      |                                                                                                                     |
|------|---------------------------------------------------------------------------------------------------------------------|
|      | On successful completion of this course, students should be able to:                                                |
| CO 1 | For a given logic sentence express it in terms of predicates, quantifiers, and logical Connectives.                 |
| CO 2 | For a given a problem, derive the solution using deductive logic and prove the solution based on logical inference. |
| CO 3 | For a given mathematical problem, classify its algebraic structure.                                                 |
| CO 4 | Evaluate Boolean functions and simplify expressions using the properties of Boolean algebra.                        |
| CO 5 | Develop the given problem as graph networks and solve with techniques of graph theory.                              |



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.  
Department of Computer Science & Engineering  
End Term Examination  
Academic Year: 2025-26 (semester I)

Class: SY

Course: Object Oriented  
Programming with Java

Course Code: OE-CS201

Date: 1/12/2025

Time: 1.30 to 3.30pm

Max. Marks: 60

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to the right indicate full marks.
- Write your answers step-by-step.

| Q.N.   | Attempt All Questions                                                                                                                                     | Marks | CO  | BT  |
|--------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----|-----|
| Q.1(a) | Define an interface? Explain with syntax and example.                                                                                                     | 5     | CO2 | BT1 |
| (b)    | Explain the working of try-catch-finally with suitable examples.                                                                                          | 5     | CO3 | BT3 |
| Q.2(a) | Write a program to show how protected members are accessible in subclasses but not in non-subclasses.                                                     | 5     | CO2 | BT2 |
| (b)    | What is a Member Inner Class? Write a program to demonstrate it.                                                                                          | 5     | CO3 | BT3 |
| Q.3(a) | Explain the entire Thread Life Cycle with a neat diagram. Discuss the methods responsible for state transitions and write a programme for multithreading. | 10    | CO3 | BT3 |
| (b)    | Write a program to demonstrate:<br><br>1. Static variable<br><br>2. Static method<br><br>3. How static members are accessed                               | 10    | CO3 | BT4 |
| Q.4(a) | What is packages in java? Explain all its types?                                                                                                          | 10    | CO3 | BT1 |
| (b)    | Explain thread synchronization in java with example? Why it is needed and what are its advantages and disadvantages?                                      | 10    | CO3 | BT2 |

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

|     |                                                                                                                                                           |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1 | Be able to understand the basics of OOP and Object-oriented approach to design software to solve real world problems based on object-oriented principles. |
| CO2 | Students can think in terms of classes, Objects, Interfaces when developing                                                                               |

|     |                                                                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     | Software.                                                                                                                                                                                   |
| CO3 | Apply object-oriented programming principles effectively in software development, identifying and leveraging the benefits of object-oriented programming Over other programming approaches. |
| CO4 | Acquire skills in testing, documenting, and preparing professional-quality software, ensuring.                                                                                              |
| CO5 | The delivery of robust and well-documented solutions.                                                                                                                                       |